

AYUSH GIRI

Mechanistic Interpretability · Trustworthy AI · LLM Reliability & Interpretability

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EDUCATION

B.Tech, Computer Science and Engineering 2020 – 2024 *Vellore Institute of Technology, Vellore*

CGPA: 8.53 / 10 · Relevant coursework: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Cloud Computing

Covid Semesters GPA: 7.21(I), 7.71(II), 8.50(II)

Post Covid Semesters GPA: 8.65(IV), 8.23(V), 9.55(VI), 9.32(VII), 9.00(VIII)

High School

Triyog High School, Nepal · GPA: 3.9/4.0

Awards: School Rank 3rd

RESEARCH EXPERIENCE

Nepal Clearing House Ltd. (NCHL) Mar 2025 – Present

Research Engineer — LLM Reliability & Interpretability

Kathmandu, Nepal

- ▶ Conducting empirical research on hallucination reduction in Retrieval-Augmented Generation (RAG) pipelines for high-stakes financial document retrieval, analyzing failure modes and where parametric knowledge overrides retrieval contextual information.
- ▶ Investigations on model alignment and factual reliability in production LLM deployments via systematic fine tuning experiments, serving infrastructure design (vLLM, Docker), and evaluation framework deployment for regulatory and compliance.
- ▶ Prototyping lightweight in-process monitors for agentic banking workflows to flag representational divergence before tool-call commitment, without external post-hoc filtering.
- ▶ Developing interpretable agent-based automation systems for banking compliance workflows, with emphasis on audit trail design and real-time monitoring as constraints on model behavior.
- ▶ Designing and researching on black box LLM interpretability techniques including activation probing (transformer-lens), behavioral analysis and confidence estimation.

Cedar Gate Technologies Dec 2024-Mar 2025

Data Trainee (Healthcare)

Kathmandu, Nepal

- ▶ Analyzed large-scale US healthcare datasets (pharmaceutical and medical claims) to extract actionable insights under HIPAA constraints; developed Data Structure Designs (DSD) for standardized data transformation and reporting.
- ▶ Built and maintained ETL pipelines mapping raw data to prod-ready formats, ensuring accuracy, consistency, and regulatory compliance across high-volume clinical data systems.

National University of Singapore (NUS) May – Jul 2023

Academic Intern — Deep Learning & Data Analytics

Singapore

- ▶ Contributed to an industry safety gear detection project applying object detection algorithms (YOLO, SSD) to CCTV footage; conducted experiments with deep learning models for low-latency, high accuracy under variable constraints.

- ▶ Worked collaboratively on optimization problems in high-noise and data augmented environments for accurate and consistent inference.

Vellore Institute of Technology (Undergrad Research) Aug 2023 - Mar 2024

Undergrad Research Assistant - Prof. Saira Banu J.

Vellore, India

- ▶ Published Harnessing ResUHybridNet with Federated Learning : A New Paradigm in Brain Tumor Segmentation for data preserving environments. Developed MediQNet, a multi-modal transformer-based medical VQA system, as final-year thesis - [Github](#) . Laid foundational skills in research methodologies, experiment designs, and reproducible DL engineering .

PUBLICATIONS

Dynamic Query Routing with Aleatoric and Epistemic Uncertainty Handling for Virtual Assistants: A Hybrid Approach in Retrieval-Augmented Generation

Giri, A. et al. · Pre-print, 2025 · [ResearchGate](#) · Under Peer-Review

Harnessing ResUHybridNet with Federated Learning: A New Paradigm in Brain Tumour Segmentation

Giri, A. et al. · Published, 2024

Privacy-preserving hybrid CNN architecture for decentralised medical image segmentation. Demonstrates reliable inference under a federated environment for data privacy and security.

CURRENT RESEARCH INTERESTS

- ▶ **Mechanistic Sources of Hallucination:** Localizing confabulations with different layers, attention circuits, and residual stream subspaces via activation patching, causal intervention and SAEs.
- ▶ **Parametric Memory vs Retrieved Context:** Understanding how attention patterns and residual-stream states mediate the competition between parametric priors and in-context evidence in RAG systems.
- ▶ **Reliability through controllable generations:** Exploring entropy-aware decoding, tractable probabilistic circuits
- ▶ **Evaluation and Detection of Faithfulness:** Conducting experiments on evaluation frameworks for hallucination detection and contextual faithfulness, LLM-as-Judge protocols, semantic entropy, and human-aligned benchmarks under controlled knowledge conflicts.
- ▶ **Alignment Science and Training Dynamics via RL:** Understanding how RLHF and RL-based fine-tuning shapes factual reliability and knowledge source preference, and whether instruction-tuned models exhibit different internal conflict resolution mechanisms than base pre-trained models.

Research Agendas

My research agenda sits in the middle of mechanistic interpretability, factual reliability, and deployment-oriented AI safety. The core thesis is that trustworthy language models require structural rather than post hoc guarantees. It is absolutely essential to understand, localize, and intervene on the internal hidden states computation in order to produce reliable outputs. This framing is motivated by building AI systems in regulated environments where hallucination is a product liability, not a benchmark metric.

My agenda spans three interconnected problem statements:

- Detecting when models hallucinate by ignoring retrieved evidence
- Understanding the internal mechanisms governing knowledge retrieval and integration
- Designing evaluation frameworks that accurately measure faithfulness under knowledge conflicts.

In the long term I want to contribute to a mechanistic theory of factual reasoning, actionable interpretability tooling and architectures for verifiable AI whose internal structure makes reliability easier to achieve by construction.

TECHNICAL SKILLS

Interpretability	TransformerLens, Activation Patching, Causal Mediation Analysis, Sparse Autoencoders, Probing Classifiers, Representation Analysis
Reliability	Uncertainty Quantification, Entropy-aware Decoding, RL and Alignment Science, Constrained Generation, RAG Architecture & Failure-mode Analysis
ML / DL	PyTorch, HuggingFace Transformers, TensorFlow, Scikit-learn, vLLM, LangChain, FAISS, Milvus, Pinecone, LORA, QLORA, Quantization, SFT, RL
Agentic / Ops	Multi-agent Frameworks, LLMOPS, W&B, Docker, FastAPI, GitLab CI, Grafana / Prometheus

CERTIFICATIONS

Data Analytics Using Deep Learning – NUS · Data Analytics with Big Data – AWS · Machine Learning Specialization – Stanford University